Two Latent Types on Arbitrary Covers

Part I: pair graphs, sign sheets and hiding polytopes
(working draft: Phases I–III; three types)

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Abstract

The lab for latent actors [1] proved its two-type results mostly on the complete pair cover of three or four variables. This companion asks the same questions for an arbitrary graph G of observed pairs on m binary variables. When do the pair tables admit a two-type explanation? Which freedoms remain? What does the set of explanations look like? This draft answers them at the pair layer, then for higher windows and for three types.

- *Existence* (theorem 2). A factorisation $c_{ij} = v_i v_j$ exists iff three conditions hold. No zero moment may join two variables that carry non-zero moments. On each component of the non-zero graph G^*, the sign cocycle must be a $\mathbb{Z}/2$ coboundary, and $\log|c|$ must lie in the image of the unsigned incidence matrix. The last two are the lab's $H^1_+ = H^1(\Gamma;\mathbb{Z}/2) \oplus \operatorname{coker} B$, extended to any graph.
- *Freedom.* $\operatorname{rank} B_G = m - b(G)$, with $b(G)$ the number of bipartite components (isolated vertices included). Odd cycles pin the magnitudes. Each bipartite component keeps a scale $\lambda^{\pm 1}$. Each component keeps its own sign, so the labels can be swapped on one component alone, invisibly to the pairs.
- *Hiding polytopes* (theorem 5). The two-type explanations fall into sign sheets. In the coordinates $y = \log t$ (the weight) and $x_k = \log \lambda_k$, each sheet is a convex polytope whose facets all have slope ± 1, two per variable: the positivity walls $a_{1i}, a_{2i} \in \{0, 1\}$. For a connected bipartite graph the sheet is a rectangle in $(\log \lambda t, \log \lambda / t)$. The generic dimension is $b(G) + 1$. The label swap identifies sheets in pairs. Positivity can empty a sheet.
- *Strata.* A coincidence $v_i = 0$ deletes vertex i from G^*. On the triangle it leaves a single edge, so the lab's hiding segment becomes a two-dimensional region.
- *Relations and syzygies* (Phase III, Stage 1, theorem 10). With triple windows the moment map is monomial, so the relations among observed moments form the toric ideal of the extended incidence matrix. Each triple either is absorbed or adds one test. Relations are disconnected fibres and syzygies are hollow loops (Hochster's formula). The syzygies are exactly the dependencies among the tests, so a test of all relations has codim degrees of freedom. On the bowtie the ideal is Hilbert–Burch: three witnesses of one latent ratio. A coincidence on a shared variable disables the tests that pass through it.
- *Higher windows and three types* (theorem 12 and propositions 13 and 14). The polynomials $s_k(r)$ are the moments of a standardised two-point latent, so higher windows obey Fibonacci gluing relations whose two-triple case is a Hankel determinant. For collinear k types the hiding set is a moment space. Two types are its lower boundary, so the lab's quadruple residual becomes a one-sided test for more types. Under pair-only disclosure

three-variable events have the same range for any number of collinear types. For three types on five variables, a single four-window is defective but two overlapping ones identify the law: S_3 descent.

- *Real signs* (theorem 15). The sign symmetries are the kernel of a parity matrix over $\mathbb{F}_2$. The lab's sign group splits into classes the relations see (their sign shadows) and odd-cycle classes no relation can see (the 2-torsion of the Smith form). Syzygies evaluated at the torus identity are the dependencies among the sign tests. A sheet lost on the latent side is a sign test gained on the data side.
- *Strata* (theorem 18). Ranks drop by an explicit rule: faces avoiding the coincident set give the latent rank, and the number of factors through a variable decides which tests survive. On pair graphs a coincidence costs one degree of freedom per extra non-bipartite lobe; a cut vertex alone costs nothing. Equal weights un-align labels that travelled through r.

Forty-one checks (CV1–CV9, SZ1–SZ8, HW1–HW8, SG1–SG8, CS1–CS8) verify every statement: on a zoo of eleven graphs, by sampling each polytope and rebuilding the law; on eight covers, by recomputing every Betti number from Hochster's formula; and on the higher windows, by recomputing every Betti table in a second computer-algebra system. The identifiability notions and walls (Phases IV–V), and the general three-type theory on arbitrary covers follow.

1 Setting

Variables are $0, \dots, m-1$. A two-type law has weight w and success probabilities $a_1, a_2 \in [0,1]^m$, with

$$\mu = wa_1 + (1-w)a_2, \qquad v = \sqrt{w(1-w)}\,(a_1 - a_2), \qquad c_{ij} = \operatorname{Cov}(X_i, X_j) = v_i v_j.$$

Following the lab, write $t = \sqrt{(1-w)/w}$, so that $w = 1/(1+t^2)$, $a_1 = \mu + vt$ and $a_2 = \mu - v/t$. The label swap is $(v,t) \mapsto (-v, 1/t)$. Observed are the single tables, which give μ, and the pair tables on the edges E of a graph $G = (V, E)$, which give c_e. On K_3 the explanations form the lab's hiding segment [1, Theorem A]. Here G is arbitrary: possibly disconnected, bipartite or not, with zero moments allowed.

Write B_G for the unsigned incidence matrix ($E \times V$, with a 1 where the vertex lies on the edge), and $b(G)$ for the number of bipartite components, an isolated vertex counting as one.

Lemma 1 (rank)**.** $\operatorname{rank} B_G = |V| - b(G)$. *The kernel is spanned, for each bipartite component with sides L, R, by the vector that is $+1$ on L, -1 on R and 0 elsewhere. An isolated vertex contributes its own coordinate.*

Proof. A vector x lies in the kernel iff $x_i = -x_j$ on every edge. Along a path x alternates in sign. On a component with an odd cycle this forces $x = 0$; on a bipartite component it forces $\pm a$ on the two sides. Check CV1 verifies the formula on 300 random graphs. □

2 Phase I: the factorisation on any pair graph

For data $c \in \mathbb{R}^E$, let N be the set of vertices on some edge with $c_e \neq 0$, and $G^* = (N, \{e : c_e \neq 0\})$ the non-zero graph. Let Z be the vertices outside N joined to N by a zero edge, and $F = V \setminus (N \cup Z)$ the *free* vertices.

Theorem 2 (factorisation)**.** *A real v with $v_i v_j = c_{ij}$ for every edge exists iff the following hold.*

(i) Zero strata. *No edge with $c_e = 0$ joins two vertices of N.*

(ii) Sign class. *On each component H of G^*, the cochain $\operatorname{sgn} c|_H \in C^1(H;\mathbb{Z}/2)$ is a coboundary. Equivalently, the product of the c_e around every cycle is positive. This is $b_1(H)$ conditions.*

(iii) Modulus class. *On each component H, $\log|c|_H \in \operatorname{im} B_H$, that is, it is orthogonal to $\ker B_H^\top$. This is $|E_H| - |V_H| + \beta_H$ equations, with $\beta_H = 1$ if H is bipartite and 0 otherwise.*

When they hold, the set of factorisations is

$$\mathcal{F}_G(c) = \prod_H \mathcal{F}_H \times \{0\}^Z \times \mathcal{S}_F,$$

where:

- $\mathcal{F}_H = \{\pm v_H\}$ *if H is not bipartite;*
- $\mathcal{F}_H = \{\sigma\lambda^\varepsilon v_H : \sigma = \pm 1,\ \lambda > 0\}$ *if H is bipartite, where $\varepsilon_i = \pm 1$ marks the side of i;*
- $\mathcal{S}_F$ *is the set of $v \in \mathbb{R}^F$ with $v_i v_j = 0$ on the zero edges inside F: the union of the coordinate subspaces of the independent sets of that graph.*

In particular the components' signs are independent, and the tangent dimension of $\prod_H \mathcal{F}_H$ is the number of bipartite components of G^*.

Proof. (i) If $v_i v_j = 0$, one of v_i, v_j is zero, but a vertex of N has a non-zero product with a neighbour, so it is not zero. This also forces $v = 0$ on Z. On G^*, write $v_i = \sigma_i e^{x_i}$. Then $v_i v_j = c_{ij}$ splits into $\sigma_i\sigma_j = \operatorname{sgn} c_{ij}$, the coboundary condition (ii) over $\mathbb{Z}/2$, and $x_i + x_j = \log|c_{ij}|$, the condition $B_H x = \log|c|$ of (iii). The two systems are independent, and lemma 1 gives their solution sets: σ up to a global sign per component, and x up to the kernel of B_H. The dimension of (ii) is the cycle rank $b_1(H)$; the codimension of $\operatorname{im} B_H$ is $|E_H| - \operatorname{rank} B_H$. On F only the zero edges constrain. Check CV2 compares the criterion with a direct non-linear solve on 66 cases (each graph in the zoo, with true, sign-corrupted and modulus-corrupted data): it agrees in all of them. Check CV3 confirms the dimension count on the zoo. □

Remark 3 (relation to the lab's descent tower)**.** On a connected pair graph, (ii) and (iii) together are Theorem M of the lab, $H^1_+ = H^1(\Gamma;\mathbb{Z}/2) \oplus \operatorname{coker} B$ [1, Theorem M]. On K_4, $\operatorname{coker} B$ is two-dimensional and spanned by the log tetrads. Three cases are new here: disconnected graphs, where the sign group becomes $(\mathbb{Z}/2)^{\#\text{components}}$; bipartite graphs, which have an extra kernel direction; and zero strata. On the 4-cycle, (iii) is the single equation $|c_{01}c_{23}| = |c_{12}c_{03}|$, an even-cycle tetrad. On odd cycles (iii) is empty.

Remark 4 (labels on one component)**.** If G^* has several components, flipping the sign of v on one component swaps the two types' labels there and only there. This is a two-type law with the same pairs but a different joint law. It is not a symmetry of the model; the windows of Part II detect it exactly when they join components with an odd count on each side (section 11).

3 Phase II: the hiding set

Fix the component signs $\sigma \in \{\pm 1\}^{\#\text{comp}(G^*)}$, a sheet. Write $y = \log t$ and $x_k = \log\lambda_k$ for the bipartite components, and choose a base factorisation v_H on each component. A vertex $i \in N$ then has $v_i = s_i a_i e^{z_i}$, with $a_i = |v_{H,i}|$, sign s_i and $z_i = \varepsilon_i x_k$ ($z_i = 0$ on rigid components).

Theorem 5 (hiding polytopes)**.** *The two-type explanations of (μ, c) on G are the points (σ, y, x) with $(y, x) \in \Pi_\sigma$, together with a value v_i for each free vertex in the interval*

$$J_i(t) = \big[\max(-\mu_i/t,\, -(1-\mu_i)t),\ \min((1-\mu_i)/t,\, \mu_i t)\big] \ni 0,$$

subject to $\mathcal{S}_F$. Here Π_σ is cut out by two inequalities per vertex $i \in N$:

$$\begin{aligned} s_i > 0: &\quad z_i + y \le \log \tfrac{1-\mu_i}{a_i}\ (a_{1i} \le 1), &\quad z_i - y \le \log \tfrac{\mu_i}{a_i}\ (a_{2i} \ge 0),\\ s_i < 0: &\quad z_i + y \le \log \tfrac{\mu_i}{a_i}\ (a_{1i} \ge 0), &\quad z_i - y \le \log \tfrac{1-\mu_i}{a_i}\ (a_{2i} \le 1). \end{aligned}$$

(1) Each Π_σ is a convex polytope, and every facet normal has the form $(\pm 1, \pm e_k)$ or $(\pm 1, 0)$: slope ± 1 in each plane (y, x_k).

(2) If G is connected and bipartite, Π is the rectangle

$$-\min_R \beta_i \le u \le \min_L \alpha_i, \qquad -\min_R \alpha_i \le w \le \min_L \beta_i,$$

where $u = x + y = \log \lambda t$, $w = x - y = \log \lambda/t$, and α_i, β_i are the right-hand sides of the first and second rows. Type 1 on L deviates by e^u and type 2 on R by e^{-u} (times the base); type 2 on L by e^w and type 1 on R by e^{-w}. The two products move independently.

(3) If every $c_e \neq 0$ and the truth has $a_{1i}, a_{2i} \in (0, 1)$ for all i, the truth lies in the interior of its sheet. The hiding set then has dimension $b(G) + 1$: one for t, one for each bipartite scale, one for each isolated vertex.

(4) The label swap maps Π_σ to $\Pi_{-\sigma}$ by $y \mapsto -y$, so there are 2^{c^*-1} distinct sheets, c^* being the number of components of G^*. Positivity can empty a sheet.

(5) On K_3 the single sheet is $y \in [\log t_-, \log t_+]$, the lab's hiding segment.

Proof. Given $v \in \mathcal{F}_G(c)$ and $t > 0$, the law (w, a_1, a_2) reproduces μ and all c_e by construction. It is a law iff $a_1, a_2 \in [0, 1]^m$. For $v_i = s_i a_i e^{z_i}$ with $s_i > 0$, the active bounds are $a_{1i} = \mu_i + a_i e^{z_i} t \le 1$ and $a_{2i} = \mu_i - a_i e^{z_i}/t \ge 0$; taking logs gives the display, and $s_i < 0$ is symmetric. Conversely, every explanation has $v \in \mathcal{F}_G(c)$ and some t, by theorem 2. For a free vertex the bounds are linear in v_i and contain 0. (1) The constraints are linear in (y, x). (2) On L, $z = x$ and the rows read $u \le \alpha_i$, $w \le \beta_i$. On R, $z = -x$ and they read $-w \le \alpha_i$, $-u \le \beta_i$. (3) Strict inequalities at the truth give an open neighbourhood. The coordinates are y, one x_k per bipartite component of G^* ($= G$ minus its isolated vertices), and one v_i per isolated vertex. These number $b(G) + 1$. (4) $v \mapsto -v$ flips every s_i, and $t \mapsto 1/t$ exchanges the two rows of each vertex. (5) K_3 has no scale, and the rows reduce to the lab's $t_- = \max(\dots)$, $t_+ = \min(\dots)$ [1, Theorem A]. Check CV5 samples every sheet of every zoo graph by hit-and-run. Each point rebuilds a law with exactly the observed singles and pairs and $a \in [0, 1]^m$, and stepping 10^{-3} beyond any facet takes the named a_{1i} or a_{2i} out of $[0, 1]$. Checks CV6–CV9 verify (2)–(5). □

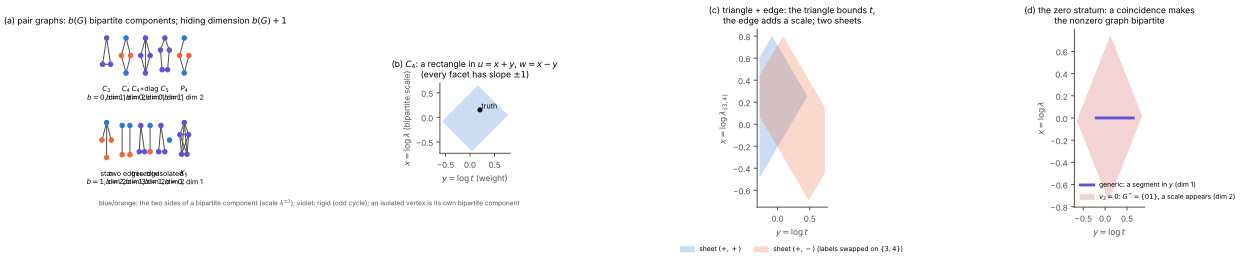


Figure 1: Phases I–II. (a) The zoo, with the two sides of each bipartite component coloured. The hiding dimension is $b(G)+1$ throughout (CV7). (b) C_4: the sheet in $(y,x)=(\log t,\log\lambda)$ is a rectangle turned 45°, a rectangle in $(\log\lambda t,\log\lambda/t)$ (example 6). (c) Triangle plus a disjoint edge: the triangle bounds t and the edge adds a scale. The two sheets differ by swapping the labels on $\{3,4\}$ only. (d) The triangle generically has a one-dimensional sheet (the lab's segment). At the coincidence $v_2=0$ the non-zero graph is the single edge 01, which is bipartite, and the sheet becomes two-dimensional.

4 Worked examples

Example 6 (C_4, every step)**.** Take $w=0.4$, $a_1=(0.8,0.3,0.7,0.25)$ and $a_2=(0.2,0.7,0.15,0.7)$, observed on the 4-cycle $01,12,23,03$.

1. *Data.* $\mu=(0.44,0.54,0.37,0.52)$ and $v=(0.2939,-0.1960,0.2694,-0.2205)$, so $c_{01}=-0.0576$, $c_{12}=-0.0528$, $c_{23}=-0.0594$ and $c_{03}=-0.0648$.

2. *Existence.* The sign product around the cycle is $(-)^4>0$, so (ii) holds. The single modulus equation is $c_{01}c_{23}/(c_{12}c_{03})=1.0000$, so (iii) holds.

3. *Factorisation.* The sides are $L=\{0,2\}$ and $R=\{1,3\}$. The minimum-norm base solution is $v_H=(0.2526,-0.2280,0.2316,-0.2565)$, and the truth is $\lambda=1.1636$ times it on L and λ^{-1} times it on R.

4. *The rectangle.* The eight rows give

$$u=\log\lambda t\in[-0.627,\,0.796],\qquad w=\log\lambda/t\in[-0.707,\,0.469].$$

   The binding walls are $a_{1,0}\le 1$ and $a_{2,3}\le 1$ for u, and $a_{2,2}\ge 0$ and $a_{1,3}\ge 0$ for w.

5. *The truth.* $t=\sqrt{0.6/0.4}=1.2247$, so $y=0.2027$, $x=\log 1.1636=0.1515$, $u=0.354$ and $w=-0.051$: inside, as fig. 1(b) shows.

So the 4-cycle leaves two dimensions free. A triple window on $\{0,1,2\}$ meets L twice and R once, so it constrains $\log|v_{012}|=x+\text{const}$, and through $s_3(r)$ also y. It removes one dimension; a second triple $\{1,2,3\}$ removes the other (section 11).

Example 7 (two disjoint edges: a sheet killed by positivity)**.** Take edges 01 and 23, with $w=0.9$, $a_1=(0.05,0.02,0.15,0.05)$ and $a_2=(0.55,0.45,0.75,0.90)$. There are two components, each bipartite, so there are $2^{2-1}=2$ sheets, in three dimensions (y,x_1,x_2). The true sheet has dimension 3. The other sheet, with the labels swapped on $\{2,3\}$, is empty. Swapping labels on one block would require that block's weight to be 0.1, so it needs $t\mapsto 1/t$ there. But t is shared, and the block's scale cannot compensate within the walls (CV9). Under less extreme weights both sheets are realised; in fig. 1(c) both sheets of the triangle-plus-edge are non-empty.

Example 8 (the zero stratum on the triangle)**.** With $a_1 = (0.8, 0.3, 0.6)$, $a_2 = (0.2, 0.7, 0.6)$ and $w = 0.4$, the variable 2 is a coincidence: $v_2 = 0$, so $c_{02} = c_{12} = 0$. Then $N = \{0, 1\}$, $Z = \{2\}$ and $G^* = \{01\}$, which is bipartite. The lab's segment becomes the two-dimensional region of fig. 1(d). The weight is no longer tied to a one-parameter curve, because nothing links the scale of v_0 to that of v_1 except their product. If instead $c_{01} = 0$ while $c_{02}, c_{12} \neq 0$, condition (i) fails and no factorisation exists (CV4).

5 Phase III, Stage 1: relations and their syzygies

Phase III is done in stages. Stage 1, this section, treats covers by pairs and triples, where the moment map is monomial. Stage 2 will add windows of size four and more, where $s_k(r)$ is no longer a monomial. Stage 3 handles real signs and parities. Stage 4 handles strata.

Algebra, reals and positivity. Before the relations, here is how they relate to the hiding polytopes. Write $\varphi : (v, r) \mapsto (m_F)_F$ for the moment map from latent to observed coordinates.

Lemma 9 (algebra, reals, positivity)**.** *(1)* Algebra (data side). *The relations of the cover generate the ideal I_A of the Zariski closure of the image of φ. Data off $V(I_A)$ have no explanation, even over $\mathbb{C}$.*

(2) Algebra (latent side). *Over data in the image with all moments non-zero, the fibre $\varphi^{-1}(c)$ is a coset of the torus $K = \{z : z^{a_j} = 1 \text{ for all } j\}$, of dimension $m + 1 - \operatorname{rank} A$. The ideal I_A describes the image; the fibre is described by $\ker A^\top$, the dual lattice.*

(3) Reals. *The real points of the fibre are $|G_2|$ translates of its positive part, one per sign symmetry (theorem 15(1)). Data in $V(I_A)$ that carry an invisible sign class have no real point in their fibre.*

(4) Positivity. *On each real translate, the constraints $a_1, a_2 \in [0, 1]^m$ cut out exactly the sheet polytope Π_σ of theorem 5, in the coordinates $(\log t, \log \lambda)$. The sign sheets are the real components of the fibre, cut by positivity. A sheet can be empty although its real translate is not.*

So the conditions are nested,

$$\text{in } V(I_A) \;\supset\; \text{real explanation} \;\supset\; \text{explanation by laws},$$

and each inclusion can be strict: the triangle with a negative product is algebraic but not real, and a dead sheet is real but not positive (example 7).

Proof. (1) The kernel of a ring map is the ideal of the closure of its image. (2) $\varphi(z) = \varphi(z')$ with all coordinates non-zero iff $(z'/z)^{a_j} = 1$ for every column. (3) $K(\mathbb{R}) = K_+ \times G_2$ (theorem 15). For (4), $a_1 = \mu + vt$ and $a_2 = \mu - v/t$ are the coordinates of theorem 5. Check BR1–BR5 (`engine/check_bridge.py`) verifies each item:

- the fibre dimensions 2, 1 and 0 on C_4, $C_4 + \{012\}$ and the bowtie;
- the bowtie's two real points;
- the liar's empty real fibre;
- on 400 random points, the sheet is exactly where positivity holds;
- the dead sheet.

□

5.1 The extended incidence matrix and the toric ideal

For a triple T, $m_T = v_T\, r$ [1, Lemma "moments"]. Observing pairs E and triples W therefore gives the monomial map

$$\varphi:\ \mathbb{Q}[c_e, m_T] \to \mathbb{Q}[v_0, \dots, v_{m-1}, r], \qquad c_{ij} \mapsto v_i v_j, \quad m_T \mapsto v_T\, r.$$

Its exponent vectors are the columns of the *extended incidence matrix* $A \in \mathbb{Z}^{(m+1)\times(|E|+|W|)}$: $e_i + e_j$ for a pair and $e_i + e_j + e_k + e_r$ for a triple. The relations among observed moments are the kernel $I_A = \ker \varphi$, the toric ideal of A [8]. A *relation* is a polynomial identity that every two-type law satisfies. A *syzygy* is a relation among relations, $\sum_j h_j g_j = 0$ with polynomial coefficients. The minimal free resolution

$$0 \leftarrow S/I_A \leftarrow S \leftarrow \bigoplus_b S(-b)^{\beta_{1,b}} \leftarrow \bigoplus_b S(-b)^{\beta_{2,b}} \leftarrow \cdots$$

records all of them, graded by $b \in \mathbb{N}A$: a monomial $v^{b'} r^{b_r}$ in the latent coordinates. We compute resolutions with Macaulay2 [4] and re-derive every multigraded Betti number independently from Hochster's formula [5], [6, Theorem 9.2]:

$$\beta_{i,b}(S/I_A) = \dim \tilde{H}_{i-1}(\Delta_b; \mathbb{Q}), \qquad \Delta_b = \{F \subseteq \text{coordinates}:\ b - \textstyle\sum_{j\in F} a_j \in \mathbb{N}A\}.$$

A face of Δ_b is a set of observed moments that can all be split off a product of latent degree b.

Theorem 10 (relations and syzygies for pairs and triples)**.** *(1)* Dimension. *The observed moments satisfy exactly the relations in I_A, with* $\operatorname{codim} I_A = n - \operatorname{rank} A$ *($n = |E| + |W|$). The latent fibre in (v, r) has dimension $m + 1 - \operatorname{rank} A$.*

(2) Pairs. *$I_A \cap \mathbb{Q}[c] = I_B$, the toric ideal of the graph. It has codimension $|E| - \operatorname{rank} B$, which is Phase I's modulus class, and is generated by the binomials of even closed walks [9, 7]: an even cycle (the 4-cycle's $c_{01}c_{23} - c_{12}c_{03}$), or two odd cycles joined by a path whose edges are counted twice.*

(3) The test ladder. *Adding one triple raises the codimension by 0 or 1. It raises it by 0 when the triple's column is independent of the others: the triple is then absorbed by r or by a bipartite scale, the lab's first probe. It raises it by 1 when the column is dependent: a new testable relation, the lab's second-triple defence [1, Corollary "second triple"].*

(4) Relations and syzygies as homology. *A minimal relation of degree b is a disconnected Δ_b ($\tilde{H}_0 \neq 0$): two ways of producing the same latent monomial from observed moments, with no observed moment in common. A first syzygy is a hollow loop ($\tilde{H}_1 \neq 0$): factorisations that are pairwise linked by relations, with no link common to all.*

(5) Syzygies are the dependencies among tests. *Let $g_1, \dots, g_N$ be minimal generators and J their Jacobian. At every model point the first syzygies, evaluated there, lie in the left kernel of J. At a smooth point* $\operatorname{rank} J = \operatorname{codim} I_A$*, and the evaluated syzygies span the left kernel, of dimension $N - \operatorname{codim} I_A$. So a Wald test of all N relations has* $\operatorname{codim} I_A$ *degrees of freedom, not N, and the syzygies say exactly which combinations are redundant. At a singular point the rank drops further.*

Proof. (1) The image of a monomial map is the toric variety of A, of dimension $\operatorname{rank} A$, and its ideal is I_A [8, Lemma 4.1]. The generic fibre has dimension (parameters) minus (image dimension).

(2) Elimination of the m_T gives the toric ideal of the columns $e_i + e_j$, the edge ring of G, whose generators are the even closed walks. (3) Adding a column raises n by one and the rank by one or zero. (4) Hochster's formula. (5) Differentiating $\sum_j h_j g_j = 0$ at a point where every g_j vanishes gives $\sum_j h_j(P)\nabla g_j(P) = 0$. At a smooth point the localised ideal is minimally generated by $\operatorname{codim} I_A$ elements, and J induces an injection $I_P/\mathfrak{m}_P I_P \to \mathfrak{m}_P/\mathfrak{m}_P^2$. Tensoring the presentation $S_P^{\beta_2} \to S_P^N \to I_P \to 0$ with the residue field is right exact, so the kernel of $k^N \to I_P/\mathfrak{m}_P I_P$, which is the left kernel of J, is the image of the evaluated syzygies. □

Checks SZ1–SZ8 verify the theorem on eight covers: $K_3 + \{012\}$; C_4 with one and with two triples; K_4 with pairs only; the bowtie $\{012\}, \{234\}$; the fan $\{012\}, \{023\}, \{034\}$; K_4 with all four triples; and the strip $\{012\}, \{234\}, \{456\}$.

- Every generator vanishes on 40 random model points and not off the model.
- All 127 multigraded Betti numbers agree with Hochster's formula. A sweep of all 614 degrees $b = Au$ with $|u| \le 4$ finds none that Macaulay2 missed.
- At random model points the Jacobian has rank $\operatorname{codim} I_A$, and the syzygies span its left kernel.

cover	n	$\operatorname{rank} A$	codim	fibre	Betti numbers $\beta_1, \beta_2, \ldots$	codim ladder
$K_3 + \{012\}$	4	4	0	0	—	$0 \to 0$
$C_4 + \{012\}$	5	4	1	1	1	$1 \to 1$
$C_4 + \{012\}, \{123\}$	6	5	1	0	1	$1 \to 1 \to 1$
K_4, pairs only	6	4	2	1	2, 1	—
bowtie	8	6	2	0	3, 2	$1 \to 1 \to 2$
fan	10	6	4	0	7, 11, 8, 3	$2 \to 2 \to 3 \to 4$
K_4, all triples	10	5	5	0	10, 20, 26, 20, 5	$2 \to 2 \to 3 \to 4 \to 5$
strip	12	8	4	0	9, 17, 12, 3	$2 \to 2 \to 3 \to 4$

All eight ideals are Cohen–Macaulay: the projective dimension equals the codimension (SZ8). For K_4 with all triples, the ten generators are the two tetrads and eight *exchange* binomials such as $c_{13}m_{023} - c_{03}m_{123}$, each trading one variable between a triple and a pair. Their codimension is 5. The five residuals the lab used at $m = 4$ (two tetrads and three differences $\hat{r}_S - \hat{r}_{S'}$ [1, Theorem G]) therefore have the right number of degrees of freedom; the other five generators are redundant, and the twenty first syzygies say how.

5.2 Worked example: the bowtie

This is the plan's five-variable example, a triangle with a second triple: windows $\{0, 1, 2\}$ and $\{2, 3, 4\}$, sharing variable 2, with all their pairs.

1. *Configuration.* There are eight coordinates, $c_{01}, c_{02}, c_{12}, c_{23}, c_{24}, c_{34}, m_{012}, m_{234}$, and six latent ones, $v_0, \ldots, v_4, r$. $\operatorname{rank} A = 6$, so the latent fibre is finite (the two triangles pin v, and one triple pins r), and the codimension is $8 - 6 = 2$.

2. *Generators.* Three, at latent degrees $v_0v_1v_2v_3v_4r$, $v_0v_1v_2^3v_3v_4r$ and $v_0v_1v_2^2v_3v_4$:

$$g_1 = c_{34}m_{012} - c_{01}m_{234}, \qquad g_2 = c_{23}c_{24}m_{012} - c_{02}c_{12}m_{234}, \qquad g_3 = c_{01}c_{23}c_{24} - c_{02}c_{12}c_{34}.$$

   g_3 is the even closed walk $0 \to 1 \to 2 \to 3 \to 4 \to 2 \to 0$. g_1 and g_2 are two forms of the second-triple test: the two triples must report the same r.

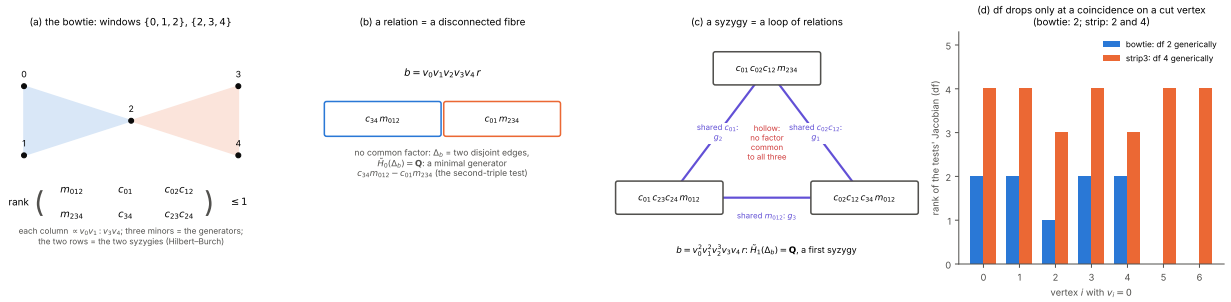


Figure 2: Phase III, Stage 1, on the bowtie. (a) The cover and its Hilbert–Burch matrix. (b) The generator g_1: the latent monomial $v_0v_1v_2v_3v_4r$ has two factorisations into observed moments with nothing in common. (c) A first syzygy: three factorisations of $v_0^2v_1^2v_2^3v_3v_4r$, pairwise linked by the generators, with no common link. (d) The tests' degrees of freedom at a coincidence on each variable: they drop only on cut variables.

3. *Hilbert–Burch.* The three are the 2×2 minors of

$$N = \begin{pmatrix} m_{012} & c_{01} & c_{02}c_{12} \\ m_{234} & c_{34} & c_{23}c_{24} \end{pmatrix},$$

whose columns all have the ratio $v_0v_1 : v_3v_4$, so they are three witnesses of one latent ratio. Since the ideal is perfect of codimension 2 with three generators, the Hilbert–Burch theorem [3, Theorem 20.15] gives the resolution $0 \to S^2 \xrightarrow{N^\top} S^3 \to S \to S/I \to 0$. The two first syzygies are the rows of N (SZ6),

$$c_{02}c_{12}\, g_1 - c_{01}\, g_2 + m_{012}\, g_3 = 0, \qquad c_{23}c_{24}\, g_1 - c_{34}\, g_2 + m_{234}\, g_3 = 0$$

Each is the expansion of a 3×3 determinant with a repeated row.

4. *As homology* (fig. 2). At $b = v_0v_1v_2v_3v_4r$, Δ_b consists of the two disjoint edges $\{c_{34}, m_{012}\}$ and $\{c_{01}, m_{234}\}$, so $\tilde{H}_0 = \mathbb{Q}$: this is g_1. At $b = v_0^2v_1^2v_2^3v_3v_4r$ the facets are $\{c_{01}, c_{02}, c_{12}, m_{234}\}$, $\{c_{01}, c_{23}, c_{24}, m_{012}\}$ and $\{c_{02}, c_{12}, c_{34}, m_{012}\}$. They meet pairwise in c_{01}, $c_{02}c_{12}$ and m_{012}, which are the links g_2, g_1 and g_3, but no moment lies in all three. So the nerve is a hollow triangle and $\tilde{H}_1 = \mathbb{Q}$: this is the first syzygy.

5. *Tests.* At a model point J has rank 2, and the two evaluated syzygies span the one-dimensional left kernel of the 3×8 Jacobian. A test of g_1, g_2, g_3 has 2 degrees of freedom. At a coincidence $v_2 = 0$ on the shared variable, the rank falls to 1: g_2 and g_3 vanish to second order, and only g_1 remains as a first-order test. At $v_i = 0$ for any other i the rank stays 2. On the strip the drops occur at the two cut variables 2 and 4, and on the fan and on K_4 (no cut variables) at none (SZ7). So a coincidence on a variable that windows share disables the tests that pass through it; this is the lab's worthless probe, seen from the relations.

6 Phase III, Stage 2: higher windows and the latent moment sequence

A window of size four or more is not monomial: $m_{0123} = v_{0123}(r^2 + 1)$. The polynomials $s_k(r)$ have a meaning, and once it is seen the relations of higher windows follow.

Lemma 11 (the standardised latent)**.** *Let $\tau = t$ on type 1 (probability w) and $\tau = -1/t$ on type 2. Then $E\tau = 0$, $E\tau^2 = 1$, $a = \mu + v\tau$, and*

$$m_S = E\prod_{i\in S} v_i\tau = v_S\, M_{|S|}, \qquad M_k = E\tau^k = s_k(r).$$

The atoms are the roots of $\tau^2 = r\tau + 1$, so $M_{k+1} = rM_k + M_{k-1}$: the s_k are the Fibonacci polynomials, the moments of a two-point law.

Proof. $wt - (1-w)/t = 0$ and $wt^2 + (1-w)/t^2 = 1$ for $w = 1/(1+t^2)$; $t\cdot(-1/t) = -1$ and $t - 1/t = r$. Check HW1 verifies $s_k = E\tau^k$ for $k \le 8$ and $m_S = v_S M_{|S|}$ on random laws. □

Observing a window observes all its marginals, so a cover is a simplicial complex K, and the coordinates are m_F for the faces with $|F| \ge 2$.

Theorem 12 (relations of higher windows)**.** *(1)* Graded pieces. *The relations form $I_K = \ker(m_F \mapsto v_F M_{|F|}(r))$, graded by the v-exponents. In degree b, I_b is the space of linear relations among the polynomials $\prod_F M_{|F|}(r)^{u_F}$, over the monomials u with $Au = b$. For pairs and triples these polynomials are powers of r, so the relations are binomials (theorem 10). From size four on, sums appear.*

(2) Fibonacci gluing. *For faces T, T' with $T \cap T' = \{i, j\}$ and $T \cup T' \in K$,*

$$c_{ij}\left(m_{T\cup T'} - m_{T\setminus j}\, m_{T'\setminus i}\right) = m_T\, m_{T'},$$

from $M_{k+l-2} = M_kM_l + M_{k-1}M_{l-1}$. For two triples the left side minus the right is $v_Tv_{T'}\,(M_4 - M_3^2 - 1)$, which is $v_Tv_{T'}$ times the Hankel determinant $\det\left(\begin{smallmatrix}1 & 0 & 1\\ 0 & 1 & M_3\\ 1 & M_3 & M_4\end{smallmatrix}\right)$.

(3) Computed ideals.

cover (faces)	*n*	*codim*	*generators*	*Betti numbers (Macaulay2 = Singular)*
the full four-variable window	*11*	*6*	*16: 2 tetrads, 8 exchanges, 6 trinomials*	*16, 41, 47, 40, 24, 5*
tetrahedron + triangle on an edge	*14*	*8*	*26, with cross-window trinomials*	*26, 92, 171, 258, 274, 161, 44, 3*
two tetrahedra on an edge	*21*	*14*	*71: 50 binomials, 21 trinomials*	*(not computed)*
the full five-variable window	*26*	*20*	*159: 104 binomials, 45 trinomials, 10 four-term*	*(not computed)*

The four-term relations come from $M_5 = r^3 + 2r$, for example $c_{01}m_{234}^2 - c_{02}c_{34}m_{1234} + m_{0234}m_{1234} - m_{234}m_{01234}$.

(4) The lab's residuals. *For the full four-variable window the codimension is* 6*, which is the rank of the lab's six residuals (two tetrads, three triple differences and the quadruple [1, Theorem G]). So those six are a local complete intersection, and the other ten generators are dependent on them through the* 41 *first syzygies.*

Proof. (1) A monomial maps to $v^{Au}\prod_F M_{|F|}^{u_F}$; different v-degrees are independent, and within one degree the relations are the linear relations among the r-polynomials. (2) The Fibonacci identity $F_{a+b} = F_{a+1}F_b + F_aF_{b-1}$ with $M_k = F_{k-1}$. Also $v_{ij}v_{T\cup T'} = v_{ij}v_{T\setminus j}v_{T'\setminus i} = v_Tv_{T'}$. (3) Macaulay2, with every Betti table recomputed by Singular's `sres` and `minres` [13] (HW2, HW4). Each of the six trinomials agrees with a gluing relation modulo the tetrads, and all of them vanish exactly in rational arithmetic on model points (HW3). (4) The Jacobian of the lab's residuals at a generic truth has rank 6 (HW2), and (5) of theorem 10 applies to any ideal at smooth points. □

7 Three types: the proof of concept ($k = 3$, $m = 5$)

Collinear types. If all k types lie on one line, $a_j = \mu + v\tau_j$, the pair moments have rank one, and lemma 11 holds with a k-atom latent τ of mean 0 and variance 1. Positivity, $0 \le \mu_i + v_i\tau \le 1$ for every i, confines τ to an interval $I = [\tau_-, \tau_+]$; the two-type walls are its ends.

Proposition 13 (collinear k types)**.** *(1) Given the pairs, the hiding set is the set of k-atom laws on I with $M_1 = 0$ and $M_2 = 1$. It has dimension $2k-3$: 1 for two types, 3 for three.*

(2) A window of size j reveals M_j. The sizes $3, 4, \dots, 2k-1$ each remove one dimension (Prony), so three types need windows of sizes 3, 4 and 5. On $m = 5$ the fibre dimension goes $3 \to 2 \to 1 \to 0$ (HW5).

(3) Two types are extreme. *Over all laws on I with $M_1 = 0$ and $M_2 = 1$, the range of M_3 is exactly the two-type segment: the extremes are two-point laws with one atom at an end of I, the principal representations of the moment problem [11, 12]. So under pair-only disclosure a three-variable event has the same range for any number of collinear types as for two. This makes Theorem A of the lab robust to the number of types.*

(4) Two types are a boundary. *$M_4 \ge M_3^2 + 1$, with equality iff τ has two atoms. Two types are the lower boundary of the (M_3, M_4) moment space, and the gluing trinomial of theorem 12(2) is its equation. Since the trinomial equals $v_Tv_{T'}(M_4 - M_3^2 - 1)$ and the sign of $v_Tv_{T'}$ is known from the pairs, the lab's quadruple residual becomes a* one-sided *test for more than two collinear types.*

Proof. (1) k atoms and k weights, less one normalisation and two moment conditions. (2) Moments up to $2k-1$ determine a generic k-atom law. (3) For a linear functional of M_3 under the three equality constraints, a basic optimal solution of the moment linear programme has at most three atoms; the principal representations of (M_0, M_1, M_2) on an interval are two-point laws. Check HW6 solves the programme on a grid of 6001 support points: the range of M_3 is $[-0.7626, 0.5731]$, equal to the two-type segment to 10^{-5}. (4) The Hankel matrix of a law is positive semidefinite, and it is singular iff the law has at most two atoms. Given $M_3 = -0.408$, M_4 is pinned at 1.1667 for two types and ranges over [1.1667, 1.8900] for three or more (HW6). □

General three types. Without collinearity, $c = VV^\top$ off the diagonal with $V \in \mathbb{R}^{5\times 2}$ (the types span a plane), and the freedom of V is $O(2)$. This is the continuous analogue of the two-type sign group $O(1) = \{\pm 1\}$.

Proposition 14 (three types on five variables; computed)**.** *(1)* Pairs. *The only relation among the ten pair moments is the pentad, a single generator of degree 5 with 12 terms [2]. It vanishes on three-type pair moments and not on random ones. The image has rank 14 of 17, so the hiding set has dimension 3.*

(2) The window ladder. *Adding triples gives ranks* $15, 16, 16, 17$*; four suitable triples, or all ten, already identify a generic three-type law locally. The third triple,* $\{0, 1, 4\}$*, adds nothing: rank depends on how the windows overlap, not only on how many there are. For two types, pairs and one triple suffice.*

(3) S_3 descent creates identifiability. *A single four-variable window is defective: rank* 13 *of* 14*, a curve of explanations. This is the known dimension deficiency of the three-class model on four binary variables [10]. Two windows* $\{0, 1, 2, 3\}$ *and* $\{1, 2, 3, 4\}$ *overlapping in a triple have rank* 17*. Multistart fitting of the glued data finds one solution up to label permutation (*32 *converged starts). The two local curves meet only where the labels match on the overlap, and the transition* $\pi \in S_3$ *is the unique permutation matching* $(w_j, a_j|_{\{1,2,3\}})$*. Gluing the windows identifies what neither identifies alone.*

(4) The non-abelian coincidence. *If two types have equal weight and agree on the overlap, the overlap data have stabiliser* S_2 *and the rank falls to* 14*: the transition is ambiguous and a continuum of glued laws appears. This is the three-type analogue of example 8, where a coincidence created a scale.*

Decisions. Figure 3(c) compares what a decision can rely on when the truth is the two-type law of the lab's worked example, for E_3 (all of $0, 1, 2$ harmed; truth 0.0130) and E_4 (all of $0, \ldots, 3$; truth 0.0086).

- Under pairs only, E_3 ranges over $[0.0068, 0.0153]$ for two types, the same interval for collinear three types (by proposition 13(3)), and at least $[0.0013, 0.0160]$ for general three types. Non-collinear explanations give each variable its own second latent direction.
- With the triple $\{0, 1, 2\}$, E_3 is read off directly. E_4 is then pinned at 0.0086 for two types. For collinear three types it lies in $[0.0075, 0.0086]$, exactly by linear programming; for general three types it lies in at least $[0.0068, 0.0100]$, an inner bound from constrained optimisation.

So the number of types decides the probe sizes a settled decision needs. For two types, triples settle every event. For three, an event on four variables needs a four-window. Model size is a decision-relevant assumption, and the one-sided test of proposition 13(4) is how the data can speak to it.

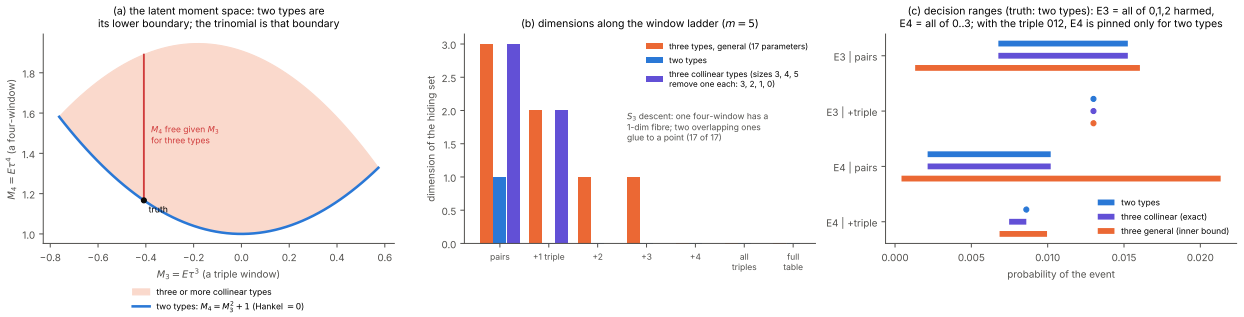


Figure 3: Higher windows and three types. (a) The moment space of the standardised latent on the positivity interval. Two types are the lower boundary $M_4 = M_3^2 + 1$, where the gluing trinomial vanishes; three or more collinear types fill the region above. (b) Hiding dimensions along the window ladder for a generic three-type truth on $m = 5$, for two types, and for collinear three types. (c) Decision ranges at a two-type truth under two-type, collinear three-type (exact) and general three-type (inner bound) explanations.

8 Phase III, Stage 3: real signs, sheets and the syzygies of sign tests

Over the reals the explanations come in sheets, and the data carry signs. This section says how the two are linked, and how the syzygies of Stage 1 govern the sign tests.

The parity matrix. Since $s_k(-r) = (-1)^k s_k(r)$, a sign change $\sigma \in \{\pm 1\}^{m+1}$ of $(v_0, \dots, v_{m-1}, r)$ preserves every observed moment iff $\prod_{i \in F} \sigma_i \, \sigma_r^{|F|} = 1$ for every face F. Over $\mathbb{F}_2$ this reads $P^\top \sigma = 0$. The *parity matrix* P has one column $1_F + (|F| \bmod 2)\, e_r$ per face; for pairs and triples it is the extended incidence matrix A reduced mod 2.

Theorem 15 (signs)**.** *(1)* Sheets. *The sign symmetries form $G_2 = \ker P^\top$, computed on the coordinates that occur in some face (a coordinate that occurs nowhere is continuous). The label swap, all signs flipped, always lies in G_2. Over generic data the real fibre is $|G_2|$ translates of one connected piece, and the sheets before positivity are $G_2/\langle \text{swap} \rangle$. For pair-only covers this is the 2^{c^*-1} of theorem 5(4).*

(2) Relative signs. *A product of signs $\prod_{i \in S} \sigma_i$ is fixed by the cover iff it is constant on G_2, which is a linear condition over $\mathbb{F}_2$. A window meeting two components in odd numbers fixes their relative sign, and one meeting them in even numbers does not. Windows that each meet only one component can also fix it, through the shared r (example 16). The odd-meeting rule stated in the earlier draft is therefore only a special case.*

(3) Visible and invisible sign classes. *For pairs and triples, the data sign patterns that pass every relation's sign test form $(L \bmod 2)^\perp$, with $2^{n - \operatorname{codim}}$ elements. The patterns realised by real explanations form $\operatorname{im} P^\top$, with $2^{\operatorname{rank}_2 A}$ elements. The quotient, the* invisible classes, *has $2^{\operatorname{rank}_{\mathbb{Q}} A - \operatorname{rank}_2 A}$ elements. Its exponent is the number of even invariant factors of the Smith form of A, the 2-torsion of* coker A. *For a pair graph,*

$$b_1(G) = \underbrace{\operatorname{codim} I_B}_{\substack{\text{visible: sign shadows of relations}}} + \underbrace{\#\{\text{non-bipartite components}\}}_{\text{invisible: the odd-cycle classes}},$$

so the lab's sign group $H^1(\Gamma;\mathbb{Z}/2)$ [1, Theorem M] splits into classes the relations see and classes no relation can see.

(4) Syzygies govern the sign tests. *A binomial generator $g_j = x^{u_j} - x^{w_j}$ imposes the sign test $\langle u_j - w_j, s\rangle \equiv 0$. Evaluated at the identity of the torus (all moments 1, which is a model point), the first syzygies are integer relations $\sum_j h_j(1)(u_j - w_j) = 0$, of rank $\#\text{gens} - \operatorname{codim}$ over $\mathbb{Q}$ and over $\mathbb{F}_2$. So exactly* codim *sign tests are independent, and the syzygies say which are redundant.*

(5) Sheets and relations trade. *Adding a face changes $\dim G_2$ by $-\Delta \operatorname{rank}_2 P$ and the number of invisible classes by $\Delta \operatorname{rank}_{\mathbb{Q}} A - \Delta \operatorname{rank}_2 A$. The four possible steps are:*

- $(1,1)$*: a freedom is absorbed and a sign fixed;*
- $(0,0)$*: a new test that carries no sign information;*
- $(1,0)$*: a freedom is absorbed but an invisible class is born, which happens exactly when an edge closes an odd cycle;*
- $(0,1)$*: a new test whose sign shadow kills a pair of sheets and turns an invisible class into a visible one.*

A sheet the latent side loses is a sign test the data side gains.

Proof. (1) σ acts by multiplication and preserves each m_F iff the parity condition holds. Over generic data with all moments non-zero, the fibre of the parametrisation is a torsor under $\{z : z^{a_j} = 1\} = K_+ \times G_2$, with K_+ the connected positive part. All ones is in G_2 because $|F| + |F|$ is even. (2) $\prod_{i\in S} \sigma_i$ is invariant iff it is trivial on G_2. (3) The sign of a monomial x^u at the sign pattern s is $\langle u, s\rangle$. The model's patterns are the image of $\sigma \mapsto P^\top \sigma$. Generators of I_A form a Markov basis, which spans the saturated lattice $L = \ker_{\mathbb{Z}} A$, so the patterns passing every test form $(L \bmod 2)^\perp$, of dimension $n - \operatorname{codim} = \operatorname{rank}_{\mathbb{Q}} A$. Over $\mathbb{F}_2$ the rank of A counts its odd invariant factors. For a graph, $\operatorname{rank}_{\mathbb{Q}} B = m - b$ and $\operatorname{rank}_2 B = m - c$, with b the bipartite and c all components; $b_1 = |E| - m + c$ and $\operatorname{codim} I_B = |E| - m + b$. (4) As in theorem 10(5), differentiate $\sum_j h_j g_j = 0$ at the smooth point 1, where $\nabla g_j(1) = u_j - w_j$ (up to the sign of g_j). The kernel of $\mathbb{Z}^{\#\text{gens}} \to L$ is saturated, so its reduction mod 2 has the same rank. (5) Rank bookkeeping. An edge raises $\operatorname{rank}_{\mathbb{Q}} B$ but not $\operatorname{rank}_2 B$ iff it joins two vertices on the same side of a bipartite component. □

Checks SG1–SG8 verify the theorem.

- On six covers with finite fibres, multistart real solving finds exactly $|G_2|$ solutions.
- On the zoo, b_1 splits as stated.
- On the Stage 1 covers, enumerating all data sign patterns gives the counts of (3): 256 pass and 128 are realised on the strip; 8 and 4 on the triangle.
- The syzygies evaluated at the identity have the ranks of (4).
- On 300 random covers, 1136 added faces fall into the four steps of (5) (522, 592, 22, 2). Every $(1,0)$ step is an edge closing an odd cycle.

Example 16 (two triangles that share only r: a reader's guide)**.** Take triangles $A = \{0,1,2\}$ and $B = \{3,4,5\}$, all six pairs, and the triple $\{0,1,2\}$.

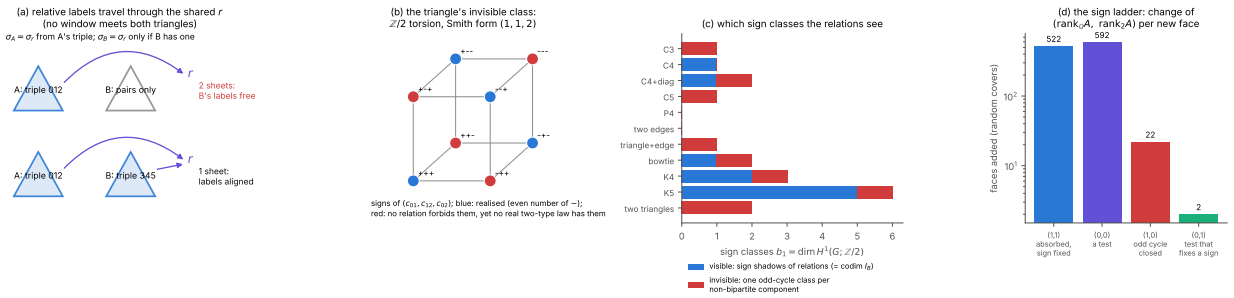


Figure 4: Real signs (theorem 15, example 16). (a) Two disjoint triangles: one triple leaves the second triangle's labels free, and a triple on each aligns them through the shared r. (b) The triangle's eight data sign patterns: four realised, four allowed by every relation but realised by nothing. (c) Over the zoo, b_1 splits into visible classes (sign shadows of relations) and invisible ones (one odd-cycle class per non-bipartite component). (d) The sign ladder over random covers: what one new face does to $(\operatorname{rank}_{\mathbb{Q}} A, \operatorname{rank}_2 A)$.

1. *Parity.* The pair columns force $\sigma_0 = \sigma_1 = \sigma_2$ and $\sigma_3 = \sigma_4 = \sigma_5$. The triple $\{0, 1, 2\}$ gives $\sigma_0\sigma_1\sigma_2\sigma_r = 1$, so $\sigma_r = \sigma_A$. Then G_2 has basis $(1, 1, 1, 0, 0, 0, 1)$ and $(0, 0, 0, 1, 1, 1, 0)$, four elements and two sheets: triangle B's labels can be swapped on their own. Real solving finds 4 solutions (SG1).

2. *Add the triple* $\{3, 4, 5\}$. Now $\sigma_r = \sigma_B$ as well, so $\sigma_A = \sigma_B$. G_2 is just the swap, there is one sheet, and 2 solutions. No window meets both triangles; the alignment travels through the weight they share.

3. *The relation it brings.* The new relation is $c_{34}c_{35}c_{45}\, m_{012}^2 = c_{01}c_{02}c_{12}\, m_{345}^2$: both triples must report the same r. Its sign test is $\operatorname{sgn}(c_{34}c_{35}c_{45}) = \operatorname{sgn}(c_{01}c_{02}c_{12})$. Before, the two triangles' odd-cycle classes were both invisible (2^2 invisible patterns). Now their difference is visible, and only their common class is not (2^1). This is the $(0, 1)$ step: one sheet pair lost on the latent side, one sign test gained on the data side.

4. *The liar, for contrast.* On a single triangle the patterns with $c_{01}c_{12}c_{02} < 0$ satisfy every relation (there are none) yet have no real explanation (fig. 4(b)). The Smith form $(1, 1, 2)$ of the incidence matrix records this class: an obstruction that lives only in the signs, never in the ideal.

Remark 17 (sheets, the tangent structure and pooling)**.** The tangent companion [14] uses these sheets. Each sheet is an object of its cone category, the layers are the walls, and a worst-case decision over a sheet is certified by the tangent cone at its minimiser. Pooling two data sets can leave the two-type domain: the pooled pair covariances can take a negative product around a triangle, which is the invisible odd-cycle class of theorem 15(3). That is the main way pooling fails globally [14, Theorem "pooling along a chord"].

9 Phase III, Stage 4: strata

In the model a zero pair moment is a coincidence ($c_{ij} = v_i v_j = 0$ forces $v_i = 0$ or $v_j = 0$). So the strata are:

- sets Z of coincident variables ($v_i = 0$ for $i \in Z$);
- equal weights ($r = 0$);
- the merge ($v = 0$).

These are the layers of the tangent companion: each stratum is a smaller cover in its own right, and the deepest one, the merge, needs a blow-up.

Theorem 18 (strata)**.** *(1)* Latent rank. *At a point with zero set Z and $r \neq 0$, the Jacobian of $(v, r) \mapsto (m_F)$ has rank*

$$\operatorname{rank} A_{K\setminus Z} + \#\{j \in Z : \text{ some face } F \text{ has } F \cap Z = \{j\}\},$$

where $A_{K\setminus Z}$ is the extended incidence matrix of the faces avoiding Z. Faces meeting Z twice vanish to second order; faces meeting it once contribute the direction e_j. The hiding dimension there is $m + 1$ minus this rank.

(2) Tests. *A generator whose A-degree in i (the number of its factors containing i) is at least 2 vanishes to second order at $v_i = 0$. The tests' degrees of freedom there are the rank of the generators of degree at most 1 in i.*

(3) Pair graphs. *At $v_i = 0$ the tests lose* $\max(0,\ \ell_i - 1)$ *degrees of freedom, where the* lobes ℓ_i *count the components of $G - i$ that are non-bipartite together with i. We prove that at least this many are lost; equality holds in every case tested.*

(4) The cut-variable conjecture, refined. *Being a cut vertex is not enough: two 4-cycles sharing a vertex lose nothing. What is lost is one degree of freedom per extra non-bipartite lobe: 1 on the bowtie, 2 on a windmill of three triangles.*

(5) Equal weights. *At $r = 0$ every odd window has $m_T = 0$ and carries no sign, so the sheet group is that of the even faces. Labels that were aligned through r come apart.*

(6) The merge. *At $v = 0$ the latent rank and every test gradient are 0, for every cover.*

Proof. (1) $\partial m_F/\partial v_j = m_F/v_j$ away from Z. At the stratum, a face with $|F \cap Z| \geq 2$ has all first derivatives zero, and a face with $F \cap Z = \{j\}$ has only $\partial_{v_j} m_F = v_{F\setminus j}\, s_{|F|}(r) \neq 0$. The remaining rows are the usual logarithmic rows of the faces avoiding Z, with r entering every higher window since $s'_k(r) \neq 0$ generically. (2) A monomial with two vanishing factors has zero gradient. A binomial is homogeneous in the A-grading, so both of its monomials have the same degree in i. (3) Two odd cycles in different lobes are joined only through i, so each figure-eight relation between lobes visits i twice (A-degree 2). There are $\ell_i - 1$ independent ones, all vanishing to second order by (2). That fewer are lost is not proved in general; checks CS4 confirm equality on 109 random graphs and four named ones. (5) At $r = 0$, $m_T = v_T s_{|T|}(0) = 0$ for odd $|T|$, so those faces impose no sign condition. (6) Every face meets $Z = V$ at least twice. □

Checks CS1–CS8 verify the theorem.

- The latent-rank rule holds in 294 single and 594 double coincidences on random covers.
- The degree rule holds on every generator of the random covers tested.
- Real solving confirms the equal-weight split: two disjoint triangles with their triples have 2 explanations for $r \neq 0$ and 4 at $r = 0$, while K_4 with its triples keeps 2.

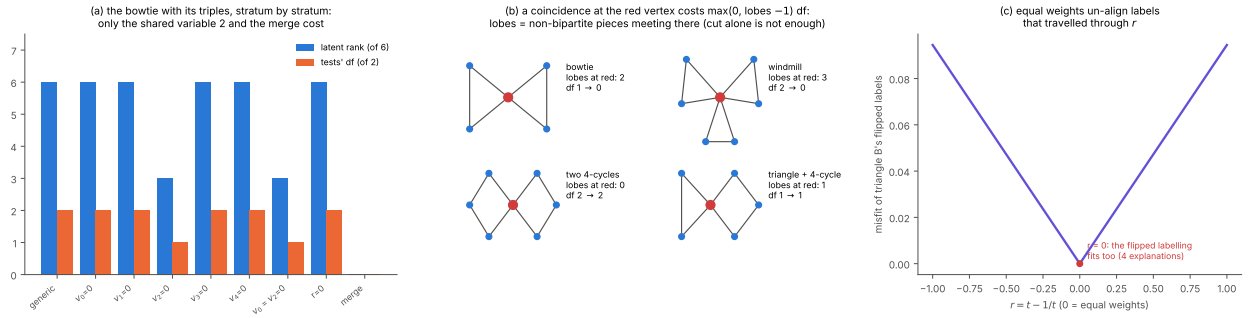


Figure 5: Strata (theorem 18). (a) The bowtie with its triples: latent rank and the tests' degrees of freedom on every stratum. (b) A coincidence at the red vertex costs $\max(0, \text{lobes} - 1)$ degrees of freedom. A cut vertex between bipartite pieces costs nothing. (c) The misfit of the explanation with triangle B's labels flipped, for two disjoint triangles with their triples. It vanishes only at $r = 0$.

Example 19 (the bowtie, stratum by stratum)**.** For the bowtie with both triples (fig. 5(a)), a coincidence on $0, 1, 3$ or 4 costs nothing: the latent rank stays 6 and the tests keep 2 degrees of freedom. A coincidence on the shared variable 2 drops the latent rank to 3, so the hiding set becomes three-dimensional. It also drops the tests to 1: the closed walk through 2 and one form of the second-triple test vanish to second order, and only $c_{34}m_{012} = c_{01}m_{234}$ survives. At $r = 0$ nothing is lost here, because the two triangles share variable 2 and are aligned by the pairs. Two disjoint triangles aligned only through r would split (fig. 5(c)). At the merge everything is lost. This is the lab's worthless probe and its $n^{-1/4}$ tempo, located exactly.

10 Where the two-type theory on covers now stands

Phase III is complete. The table records, for any cover (a simplicial complex of observed windows), what each layer of structure is, how it is computed, and where it changes.

layer	what it is	how it is computed / where it changes
existence	factorisation of the pairs; sign class; modulus class; positivity	theorem 2: zero strata, $H^1(G;\mathbb{Z}/2)$, $\operatorname{coker} B$, the walls
continuous freedom	weight t, one scale per bipartite component, free variables	$m+1-\operatorname{rank}_{\mathbb{Q}} A$; hiding polytopes with facets of slope ± 1 (theorem 5)
relations	tests among the observed moments	toric ideal for pairs and triples (theorem 10); Fibonacci–Hankel relations for higher windows (theorem 12)
syzygies	dependencies among the tests	the tests have codim degrees of freedom; homology of Δ_b; at the torus identity, dependencies among sign tests (theorem 15(4))
sheets	discrete label freedom	$G_2 = \ker P^{\top}$ over $\mathbb{F}_2$; relative signs by linear algebra; visible versus invisible classes (2-torsion of $\operatorname{coker} A$)
strata	where ranks drop	latent rank by faces avoiding Z; tests by A-degree; lobes on pair graphs; $r=0$ unaligns; the merge (theorem 18)
more types	collinear k types; three types on five variables	moment spaces, window sizes $3,\dots,2k-1$; pentad, $O(2)$, S_3 descent (propositions 13 and 14)

What remains is consolidation, not new structure:

- *Phases IV–V.* The three notions of identifiability (factorisation, parameters, law) and the walls, read off this table.
- The Betti tables of the two largest covers.
- The general three-type theory on arbitrary covers.

11 What comes next

Loose ends of Phase III. Prove equality in theorem 18(3); a lobe rule for covers with windows (where r joins lobes, as in example 16); the Betti tables of the two largest covers.

Phases IV–V. Separate factorisation, parameter and distribution identifiability; list the walls where sheets lose dimension.

Three types on arbitrary covers. Replace the sign cocycle by an $O(2)$-valued one on the pair layer, and the label sheets by S_3-torsors on the window nerve. The cocycle condition around a cycle of three windows is the first place where a genuinely non-abelian obstruction can appear.

Checks

Phases I–II are checked in `engine/check_covers.py` (module `engine/pce/covers.py`; 9/9, under two minutes). Phase III, Stage 1 is checked in `engine/check_syzygy.py` (module `engine/pce/syzygy.py`; 8/8, about thirty seconds; needs Macaulay2). Stage 2 and the three-type proof of concept are checked in `engine/check_higher.py` (modules `engine/pce/syzygy.py`,

`engine/pce/threetype.py`; 8/8, a few minutes; needs Macaulay2 and Singular). The decision ranges of fig. 3(c) for general three types come from `covers/fig_higher.py` and a constrained-optimisation run recorded with it; they are inner bounds. Stage 3 is checked in `engine/check_signs.py` (module `engine/pce/signs.py`; 8/8; needs Macaulay2), and Stage 4 in `engine/check_cover_strata.py` (module `engine/pce/cover_strata.py`; 8/8). Lemma 9 is checked in `engine/check_bridge.py` (5/5; needs Macaulay2).

CV1	rank $B_G = \|V\| - b(G)$ on 300 random graphs
CV2	existence criterion against a direct non-linear solve: 66/66 (44 factorisable, 15 sign-class and 7 modulus-class failures)
CV3	factorisation set and its tangent dimension on the zoo
CV4	zero strata: coincidence on the triangle creates a scale; zero edge inside N infeasible
CV5	every sampled point of every sheet rebuilds the observed singles and pairs; each facet is a named wall
CV6	connected bipartite graphs: rectangles in $(\log \lambda t, \log \lambda/t)$
CV7	dimension $b(G) + 1$ on all eleven zoo graphs; sheet counts
CV8	K_3 recovers the lab's segment exactly
CV9	a sheet emptied by positivity
SZ1	generators vanish on the model and not off it; codim $= n - \operatorname{rank} A$
SZ2, SZ3	all multigraded Betti numbers equal Hochster's formula; none missed up to $\|u\| \le 4$
SZ4	pair part $= I_B$, the even closed walks, codimension $\|E\| - \operatorname{rank} B$
SZ5	the test ladder: each triple adds 0 or 1
SZ6	bowtie: Hilbert–Burch, syzygies the rows of N
SZ7	syzygies span the left kernel of the tests' Jacobian; degrees of freedom drop at cut-variable coincidences
SZ8	Cohen–Macaulay; Betti tables
HW1	$s_k = E\tau^k$; $\tau^2 = r\tau + 1$; $m_S = v_S M_{\|S\|}$
HW2	the full four-window: 2 tetrads, 8 exchanges, 6 trinomials; Betti (M2 = Singular); lab residuals rank 6
HW3	gluing relations exact; trinomials = gluing relations modulo tetrads; Hankel
HW4	tetrahedron + triangle: cross-window trinomials; Betti (M2 = Singular)
HW5	collinear three types: fibre $3 \to 2 \to 1 \to 0$ with window sizes $3, 4, 5$
HW6	moment space: M_3 range = two-type segment; M_4 given M_3 an interval for three types
HW7	pentad; the three-type ladder $14, 15, 16, 16, 17$
HW8	one four-window defective; two overlapping ones identify (S_3 descent); coincidence rank 14
SG1, SG2	sheets $= \|G_2\|$ real solutions; relative labels through r
SG3	the $\mathbb{F}_2$ criterion versus the odd-meeting rule
SG4, SG5	invisible classes = even Smith invariants; b_1 = visible + invisible
SG6	data sign patterns: passing $2^{n-\operatorname{codim}}$, realised $2^{\operatorname{rank}_2 A}$
SG7	syzygies at the torus identity: the dependencies among sign tests
SG8	the sign ladder
BR1–BR5	lemma 9: image ideal, torus fibre, real translates, positive part = sheet; the nesting; walls on a sheet
CS1, CS2	latent rank at single and double coincidences
CS3	tests: degree ≥ 2 generators vanish to second order
CS4, CS5	pair graphs: $\max(0, \text{lobes} - 1)$; the cut-variable conjecture refined
CS6	equal weights un-align labels aligned through r
CS7	the merge: rank 0, df 0
CS8	the bowtie's strata table

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